

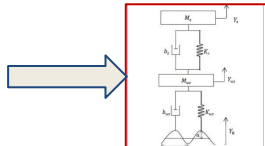
Abstract

Accurately modeling vehicle suspension is essential for ride comfort evaluation, structural design, and real-time control. Classical quarter-car models assume linear spring and damper behavior, but real suspension systems exhibit nonlinearities that these models cannot capture. This project presents a data-driven approach using a Gated Recurrent Unit (GRU) recurrent neural network to learn the input-output dynamics of a Quanser active suspension system directly from experimental data, bypassing the need for explicit system identification or knowledge of physical parameters.

The model accepts a sliding window of ISO 8608 road profile elevation as input and simultaneously predicts vehicle body and tire displacement. Training was performed across all five ISO 8608 roughness classes (A–E), using 100,000 data points per class sampled at 1 kHz. The GRU achieved a correlation of ~ 0.999 across all road classes, with body displacement RMSE ranging from 0.0204 mm (Class A) to 0.0709 mm (Class E).

Problem Definition

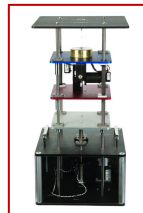
- Predicting suspension response to road excitation is central to ride comfort evaluation, structural design and real-time control
- Classical quarter-car models require known physical parameters (stiffness, damping, mass) and assume linear behavior
- Real suspension systems exhibit complex nonlinear dynamics
- Goal:** Predict body and tire displacement simultaneously from road profile using a data-driven neural network without explicit system identification
- Road inputs follow the ISO 8608 standard
- Experimental data collected from Quanser Active Suspension bench system for controlled measurements



Materials and Methods

Dataset

- Quanser Active Suspension benchmark dataset (Zenodo)
- ISO 8608 road profiles - 5 roughness classes (A–E) with multisine signals
- Inputs applied to physical hardware, body and tire displacements recorded directly
- 100,000 samples at 1 kHz - 2 km road at 20 m/s
- Global Z-score scaling

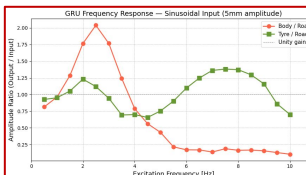
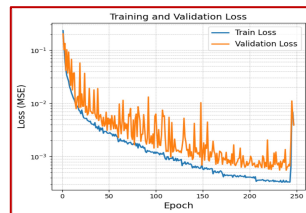


Network Architecture

- GRU chosen because suspension response is temporally and contextually dependent
- Two stacked GRU layers for sequential road profile processing
- Three-layer fully connected MLP decoder
- Sequence length of 1000 samples (1 second) with stride of 5 samples
- Trained for 300 epoch, stopped early when validation loss spiked
- Trained jointly on all five classes
- Libraries: PyTorch, SciPy, NumPy, Matplotlib, scikit-learn

Model Validation

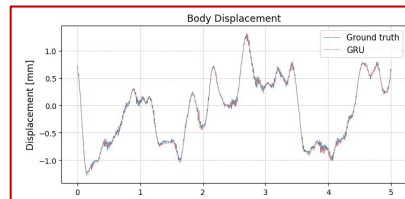
- Frequency response validated by sweeping sinusoidal inputs (0.5-10 Hz) through the trained model
- Body resonance peak ~ 2.5 Hz and high frequency attenuation consistent with physical dynamics
- Tire resonance band 7-8 Hz consistent with tire-hop frequency behavior
- Best validation loss: $5.803e-04$



Results

Class	Body RMSE (mm)	Tire RMSE (mm)	Body Correlation	Tire Correlation
A	0.0204	0.0142	0.9970	0.9983
B	0.0234	0.0207	0.9993	0.9995
C	0.0293	0.0360	0.9996	0.9996
D	0.0482	0.0623	0.9998	0.9997
E	0.0709	0.1002	0.9999	0.9998

GRU achieves correlation ≥ 0.997 across all road classes, with body RMSE ranging from 0.0204 mm (Class A) to 0.0709 mm (Class E)



Key Takeaways

- GRU accurately predicts body and tire displacement across all road classes with high correlation and without prior parameter knowledge
- Frequency response testing confirms realistic dynamics are captured
- Training on all classes resulted in generalized model for various scenarios
- Limitation: Negative prediction bias for smaller amplitudes

Future Direction and Applications

- Implement physics-informed loss functions (Energy balance), although real nonlinearities may conflict with simplified physics assumptions
- Extended training to real-world road data rather than synthetic profiles
- Deploy in active suspension ECUs for real time displacement prediction and parameter adjustment for comfort or performance

Acknowledgments

I would like to thank Professor Vlassis for his guidance and instruction in this course. The experimental data used in this study, consisting of 100,000 data points per ISO 8608 roughness class sampled at 1 kHz, was obtained from the Quanser active suspension dataset hosted on Zenodo (<https://zenodo.org/records/17232645>).